HW8

Group 12 - Shir, Omer & Ido

09/06/26

Group:

Group members (Name, ID, email)

- Ido Ravid, 207198177, ido.ravid@mail.huji.ac.il
- Omer Sutovsky, 318735107 ,omer.sutovsky@mail.huji.ac.il
- Shir Tsarfaty, 207718008, shir.tsarfaty@mail.huji.ac.il

Paths and Data

```
if (!require('data.table')) { install.packages('data.table'); library('data.table') }
```

```
## Loading required package: data.table
```

```
if (!require('splines'))    { install.packages('splines');    library('splines') }
```

```
## Loading required package: splines
```

```
source("../utils.R")
source("../lab5/lab5_utils.R")

healthy_file <- "../lab2/data/TCGA-13-0723-10B_lib2_all_chr1.forward"
tumor_file   <- "../lab7/data/TCGA-13-0723-01A_lib2_all_chr1.forward"
chr1_file    <- "../lab1/data/chr1_str.rda"
```

Introduction

In this lab we compare the main statistical models we came up with throughout the semester- both single-sample and two-sample models, and GC-corrected and non-corrected variants.

Incorporating both GC content and two-samples should help us capture the majority of the variance caused by technical effects, and thus predict copy number efficiently.

As mentioned in the lab description, we focus on two specific regions. with ~0.4 GC content and use 2.5kb windows.

The four methods are:

- **(a)** No correction — raw read counts, median normalized to 1.
- **(b)** Single-sample GC correction — divide by the fitted GC spline, then median-normalize (lab 6/7 estimator).
- **(c)** Two-sample correction without GC — divide the tumor estimate by the healthy estimate, without any GC correction.
- **(d)** Two-sample correction after GC — divide the GC-corrected tumor by the GC-corrected healthy.

Question A

```
healthy_reads <- fread(healthy_file, col.names = c("Chr", "Loc", "Length"))
tumor_reads   <- fread(tumor_file,   col.names = c("Chr", "Loc", "Length"))
load(chr1_file)
```

```
BIN  <- 2500
BEG  <- 5000001L
END  <- 100000000L

df_h <- bin_data(healthy_reads, chr1, BEG, END, BIN)
df_t <- bin_data(tumor_reads,   chr1, BEG, END, BIN)

# genomic midpoint of each bin
n_bins      <- nrow(df_h)
df_h$pos    <- BEG + (seq_len(n_bins) - 1) * BIN + BIN / 2
df_t$pos    <- df_h$pos

cat(sprintf("Bins: %d  |  region: %.0f - %.0f Mb\n",
            n_bins, BEG / 1e6, (BEG + n_bins * BIN) / 1e6))
```

```
## Bins: 38000  |  region: 5 - 100 Mb
```

```
# Fit GC splines on non-zero bins (same as lab 7)
fit_h <- fit_gc_spline(df_h[df_h$reads > 0, ])
fit_t <- fit_gc_spline(df_t[df_t$reads > 0, ])
cat(sprintf("R²  Healthy: %.3f | Tumor: %.3f\n",
            summary(fit_h)$r.squared, summary(fit_t)$r.squared))
```

```
## R²  Healthy: 0.683 | Tumor: 0.521
```

```
eps <- 1  # stabilizer as discussed in class.

# GC-based expected coverage (floored at 0)
fhat_h <- pmax(predict(fit_h, data.frame(gc = df_h$gc)), 0)
```

```
## Warning in bs(gc, degree = 3L, knots = c(`25%` = 0.3748, `50%` = 0.4176, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

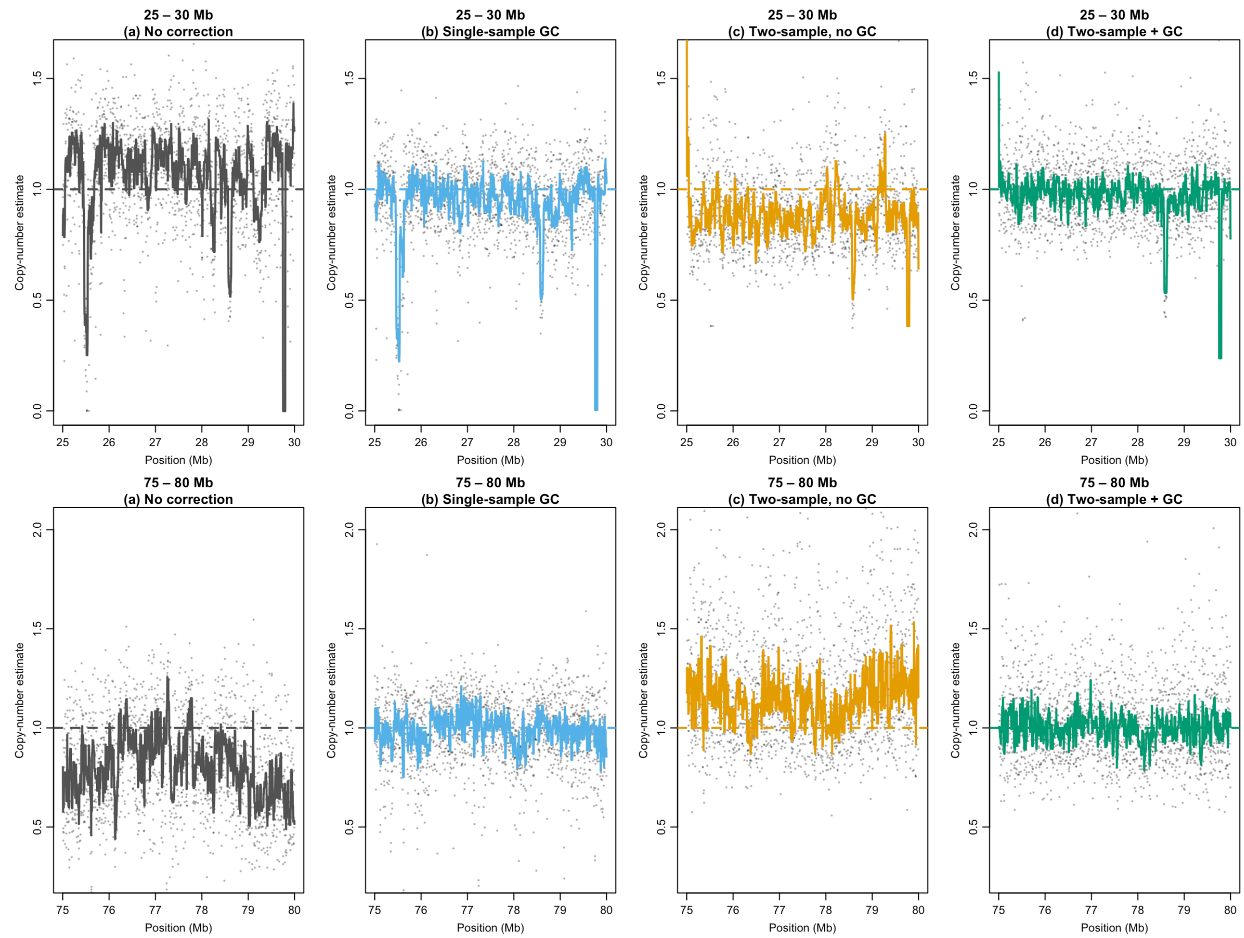
```
fhat_t <- pmax(predict(fit_t, data.frame(gc = df_t$gc)), 0)
```

```
## Warning in bs(gc, degree = 3L, knots = c(`25%` = 0.3748, `50%` = 0.4176, : some
## 'x' values beyond boundary knots may cause ill-conditioned bases
```

```
# (a) no correction: raw, median-normalized
a_h <- df_h$reads / median(df_h$reads[df_h$reads > 0])
a_t <- df_t$reads / median(df_t$reads[df_t$reads > 0])

# (b) single-sample GC correction (lab-6 plug-in estimator)
b_raw_h <- (df_h$reads + eps) / (fhat_h + eps)
b_raw_t <- (df_t$reads + eps) / (fhat_t + eps)
b_h <- b_raw_h / median(b_raw_h[df_h$reads > 0])
b_t <- b_raw_t / median(b_raw_t[df_t$reads > 0])
```

```r
# (c) two-sample, no GC: tumor / healthy ratio after median normalization
c_hat <- (df_t$reads + eps) / (df_h$reads + eps) *
          median(df_h$reads[df_h$reads > 0]) /
          median(df_t$reads[df_t$reads > 0])

# (d) two-sample after GC: GC-corrected tumor / GC-corrected healthy
d_hat <- b_t / b_h
```

```r
regions <- list(
  list(label = "25 - 30 Mb", lo = 25e6, hi = 30e6),
  list(label = "75 - 80 Mb", lo = 75e6, hi = 80e6)
)

methods <- list(
  list(y = a_t,  col = "#555555", lbl = "(a) No correction"),
  list(y = b_t,  col = "#56B4E9", lbl = "(b) Single-sample GC"),
  list(y = c_hat, col = "#E69F00", lbl = "(c) Two-sample, no GC"),
  list(y = d_hat, col = "#009E73", lbl = "(d) Two-sample + GC")
)

par(mfrow = c(2, 4), mar = c(3.5, 3.5, 2.5, 1), mgp = c(2, 0.7, 0))

for (reg in regions) {
  idx <- df_h$pos >= reg$lo & df_h$pos <= reg$hi
  x   <- df_h$pos[idx] / 1e6

  # shared y-axis across all four methods for this region
  all_vals <- unlist(lapply(methods, function(m) m$y[idx]))
  ylim_use <- quantile(all_vals, c(0.005, 0.995), na.rm = TRUE)
  ylim_use[1] <- max(0, ylim_use[1] - 0.1)
  ylim_use[2] <- ylim_use[2] + 0.1

  for (m in methods) {
    y <- m$y[idx]

    plot(x, y,
         pch = 20, cex = 0.35, col = rgb(0, 0, 0, 0.25),
         xlab = "Position (Mb)", ylab = "Copy-number estimate",
         main = sprintf("%s\n%s", reg$label, m$lbl),
         ylim = ylim_use)
    abline(h = 1, col = m$col, lwd = 2, lty = 2)

    # running median to show trend
    ord  <- order(x)
    smth <- runmed(y[ord], k = 11)
    lines(x[ord], smth, col = m$col, lwd = 2)
  }
}
```

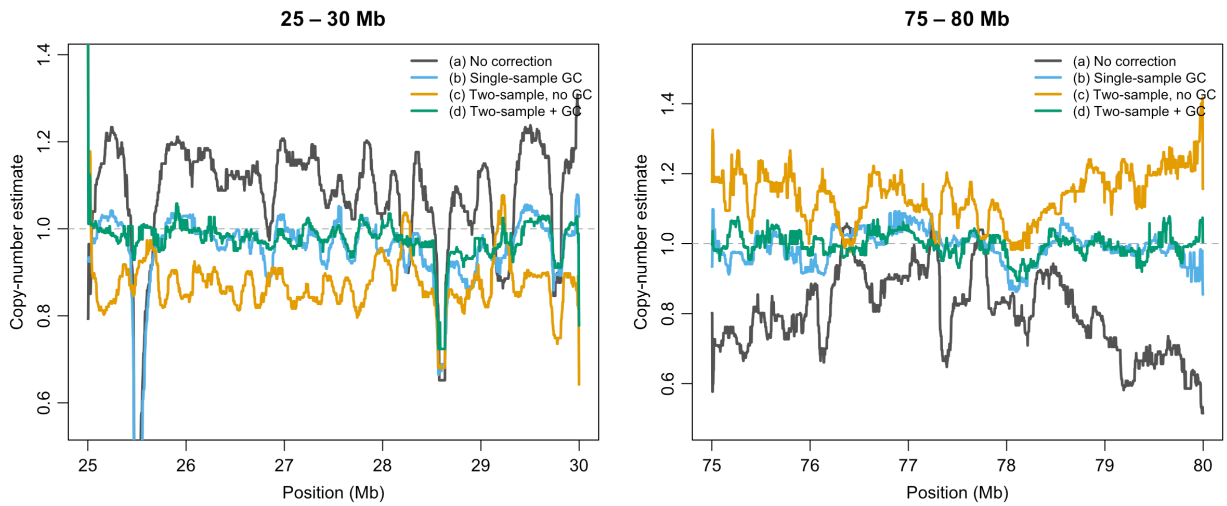


Each panel shows a dashed reference at CN = 1 and a running median trend line.

In the 25–30 Mb region, a dip below 1 near 28.5 Mb becomes visible after GC correction (b, d) and is sharpest in the two-sample methods, which also cancel shared technical variation between the libraries. In the 75–80 Mb region, uncorrected estimates are noisier and off-center; GC correction brings them back toward 1.

```
method_cols  <- c("(a)" = "#555555", "(b)" = "#56B4E9",
                  "(c)" = "#E69F00", "(d)" = "#009E73")
method_names <- c("(a) No correction", "(b) Single-sample GC",
                  "(c) Two-sample, no GC", "(d) Two-sample + GC")
all_y <- list(a_t, b_t, c_hat, d_hat)

par(mfrow = c(1, 2), mar = c(3.5, 3.5, 2.5, 1), mgp = c(2, 0.7, 0))

for (reg in regions) {
  idx <- df_h$pos >= reg$lo & df_h$pos <= reg$hi
  x   <- df_h$pos[idx] / 1e6

  # y-axis range: clip at 5th/95th pct across all four methods combined
  all_vals <- unlist(lapply(all_y, function(y) y[idx]))
  ylim_use <- quantile(all_vals, c(0.05, 0.95), na.rm = TRUE)
  ylim_use <- c(max(0, ylim_use[1] - 0.1), ylim_use[2] + 0.1)

  plot(NA, xlim = range(x), ylim = ylim_use,
       xlab = "Position (Mb)", ylab = "Copy-number estimate",
```

```
      main = reg$label)
  abline(h = 1, col = "grey70", lwd = 1, lty = 2)

  for (i in seq_along(all_y)) {
    y    <- all_y[[i]][idx]
    ord  <- order(x)
    smth <- runmed(y[ord], k = 51)
    lines(x[ord], smth, col = method_cols[i], lwd = 2.5)
  }

  legend("topright", legend = method_names,
         col = method_cols, lwd = 2.5, bty = "n", cex = 0.8)
}
```



The overlay graph allows us to compare more directly -In the 75-80 MB region, the GC-corrected graphs generally create more conservative estimates, while the the Two sample generaly estimates the CN to be higher then 1, and the uncorrected estimator generates lower estimates. in the 25-30 Mb region, the GC-normalized graph are still more conservatrive but the non-normalized estimators switch roles in overestimation/underestimation.

Question B

We use a t-test to quantify how well each method separates the CN loss event from the surrounding background. The t-statistic measures the difference between the event mean and the background mean, scaled by the combined within-group noise:

$$t = \frac{\bar{a}_{\text{bg}} - \bar{a}_{\text{event}}}{\sqrt{s^2_{\text{bg}}/n_{\text{bg}} + s^2_{\text{event}}/n_{\text{event}}}}$$

A larger t means the loss signal is more clearly detectable above the noise floor — both a bigger mean shift and lower within-region variance contribute.

- **Event region**: 28.0–29.0 Mb (the known CN loss near 28.5 Mb, $a_k < 1$).
- **Background**: flanking windows 25–28 Mb and 29–30 Mb, expected CN = 1.

```r
event_mask <- df_h$pos >= 28.0e6 & df_h$pos <  29.0e6
bg_mask    <- (df_h$pos >= 25.0e6 & df_h$pos < 28.0e6) |
              (df_h$pos >= 29.0e6 & df_h$pos <= 30.0e6)

welch_t <- function(y) {
  y_clean <- pmin(y, 5)
  ev <- y_clean[event_mask]; ev <- ev[is.finite(ev)]
  bg <- y_clean[bg_mask];   bg <- bg[is.finite(bg)]
  tt <- t.test(bg, ev, alternative = "greater")
  data.frame(mean_bg    = round(mean(bg), 3),
             mean_event = round(mean(ev), 3),
             t_stat     = round(tt$statistic, 2),
             p_value    = signif(tt$p.value, 3))
}

results <- do.call(rbind, lapply(
  list("(a) No correction"     = a_t,
       "(b) Single-sample GC"  = b_t,
       "(c) Two-sample, no GC" = c_hat,
       "(d) Two-sample + GC"   = d_hat),
  welch_t))

knitr::kable(results,
             col.names = c("Mean CN (background)", "Mean CN (event)", "t-statistic",
              ↪  "p-value"),
             caption   = "Welch t-test: background vs. event region (~28.5 Mb CN loss)")
```

Table 1: Welch t-test: background vs. event region (~28.5 Mb CN loss)

	Mean CN (background)	Mean CN (event)	t-statistic	p-value
(a) No correction	1.065	1.016	3.62	1.60e-04
(b) Single-sample GC	0.953	0.907	4.34	8.10e-06
(c) Two-sample, no GC	0.890	0.887	0.38	3.52e-01
(d) Two-sample + GC	0.990	0.943	5.03	3.00e-07

Method (d) gives the highest t-statistic, confirming it best separates the loss from background noise. Method (b) is second best, proving that the GC correction alone already helps. Method (a) shows weaker separation, and method (c) pretty much fails- the raw ratio flattens the entire region, and the event is indistinguishable from the background.

Question C

Method	Advantages	Disadvantages
(b) Single-sample GC	1. Only one sample needed; 2. adresses the main technical effect	1. doesn't capture effects in this specific sample 2. Splines don't estimate well in low/high GC areas.

Method	Advantages	Disadvantages
(c) Two-sample, no GC	1. Simpler model statistically 2. Sample-specific effects are addressed	1. Does not capture the GC effect, and may make it become noisier. 2. Needs 2 samples
(d) Two-sample + GC	1. Has both GC effect and sample-specific effects	1. More complicated statistical model 2. GC estimator may cause extreme values (as we mentioned in class) 3. Needs 2 samples

---

External sources statement

We used AI (Claude) to:

1. Offer alternatives to the statistic in B. In the end went with our own suggestion.
2. Format C in a table, Format equation for B.
3. Correct model syntax in A.